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    "Split s = \"Hi there Sam!\" into a list."
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  "s.split()"
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  "Given: planet = \"Earth\", diameter = 12742,use . format()to print:\n",
  "The diameter of Earth is 12742 kilometers\n"
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   ]
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```
 ],
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  "diameter = 12742\n",
  "\n",
  "print(\"The diameter of {} is {} kilometers.\".format(planet, diameter))"
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  "# Task 04\n",
  " Use indexing to grab \"hello\" from the given nested list: \n",
  " lst = [1, 2, [3, 4], [5, [100, 200, ['hello']], 23, 11], 1, 7]"
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  "lst[3][1][2][0]"
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    "Use indexing to grab \"hello\" from the nested dictionary   Given: d = {'k1': [1, 2, 3, {'tricky': ['oh', 'man', 'inception', {'target': [1, 2, 3, 'hello']}]}]}"
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    "d = {'k1': [1, 2, 3, {'tricky': ['oh', 'man', 'inception', {'target': [1, 2, 3, 'hello']}]}]}\n",
    "\n",
    "d['k1'][3]['tricky'][3]['target'][3]"
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    "What is the main difference between a tuple and a list?\n",
    "\n",
    "# Ans:\n",
    "Lists are mutable (their elements can be modified, added, or changed after creation).\n",
    "Tuples are immutable (their elements cannot be modified, changed, or reassigned once created)."
```

```
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  " Function: grab the domain from \"user@domain.com\""
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  "# Example Usage:\n",
  "domainGet('user@domain.com')"
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  "def findDog(st):\n",
  "    return 'dog' in st.lower()\n",
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  "# Example Usage:\n",
  "findDog('Is there a Dog here?')"
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  "Function: count how many times \"dog\" occurs in a string (case-insensitive)"
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   "\n",
   "# Example Usage:\n",
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     "No Ticket\n",
     "Big Ticket\n",
```

```
    "Small Ticket\n"
   ]
  }
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  "def caught_speeding(speed, is_birthday):\n",
  "    # Adjust speed threshold if it's the driver's birthday\n",
  "    if is_birthday:\n",
  "        speed = speed - 5  # Effectively adds 5 mph allowance\n",
  "        \n",
  "    # Evaluate speed limits\n",
  "    if speed <= 60:\n",
  "        return \"No Ticket\"\n",
  "    elif speed <= 80:\n",
  "        return \"Small Ticket\"\n",
  "    else:\n",
  "        return \"Big Ticket\"\n",
  "        # Test Case 1: Speed is 60, not a birthday\n",
  "print(caught_speeding(60, False)) \n",
  "# Test Case 2: Speed is 65, on birthday\n",
  "print(caught_speeding(65, False))  \n",
  "# Test Case 2: Speed is 65, on birthday\n",
  "print(caught_speeding(65, True))  \n",
  "# Test Case 3: Speed is 81, on birthday\n",
  "print(caught_speeding(81, False))  \n",
  "# Test Case 4: Speed is 85, not a birthday\n",
  "print(caught_speeding(85, True))\n"
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